7 Scalar and vector quantities

What students need to learn	Notes
A The addition and subtraction of coplanar vectors and the multiplication of a vector by a scalar	Knowledge of the fact that if $\alpha_1\mathbf{a} + \beta_1\mathbf{b} = \alpha_2\mathbf{a} + \beta_2\mathbf{b}$, where $\mathbf{a}$ and $\mathbf{b}$ are non-parallel vectors, then $\alpha_1 = \alpha_2$ and $\beta_1 = \beta_2$, is expected.
B Components and resolved parts of a vector	Use of the vectors $\mathbf{i}$ and $\mathbf{j}$ will be expected.
C Magnitude of a vector	
D Position vector	$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \mathbf{b} - \mathbf{a}$
E Unit vector	
F Use of vectors to establish simple properties of geometrical figures	The 'simple properties' will, in general, involve collinearity, parallel lines and concurrency. Position vector of a point dividing the line AB in the ratio $m : n$ is expected.